UML Class Diagram

BinarySearchTree, TreeNode and the client code (milestone4.cpp). '+' public, '-' private.

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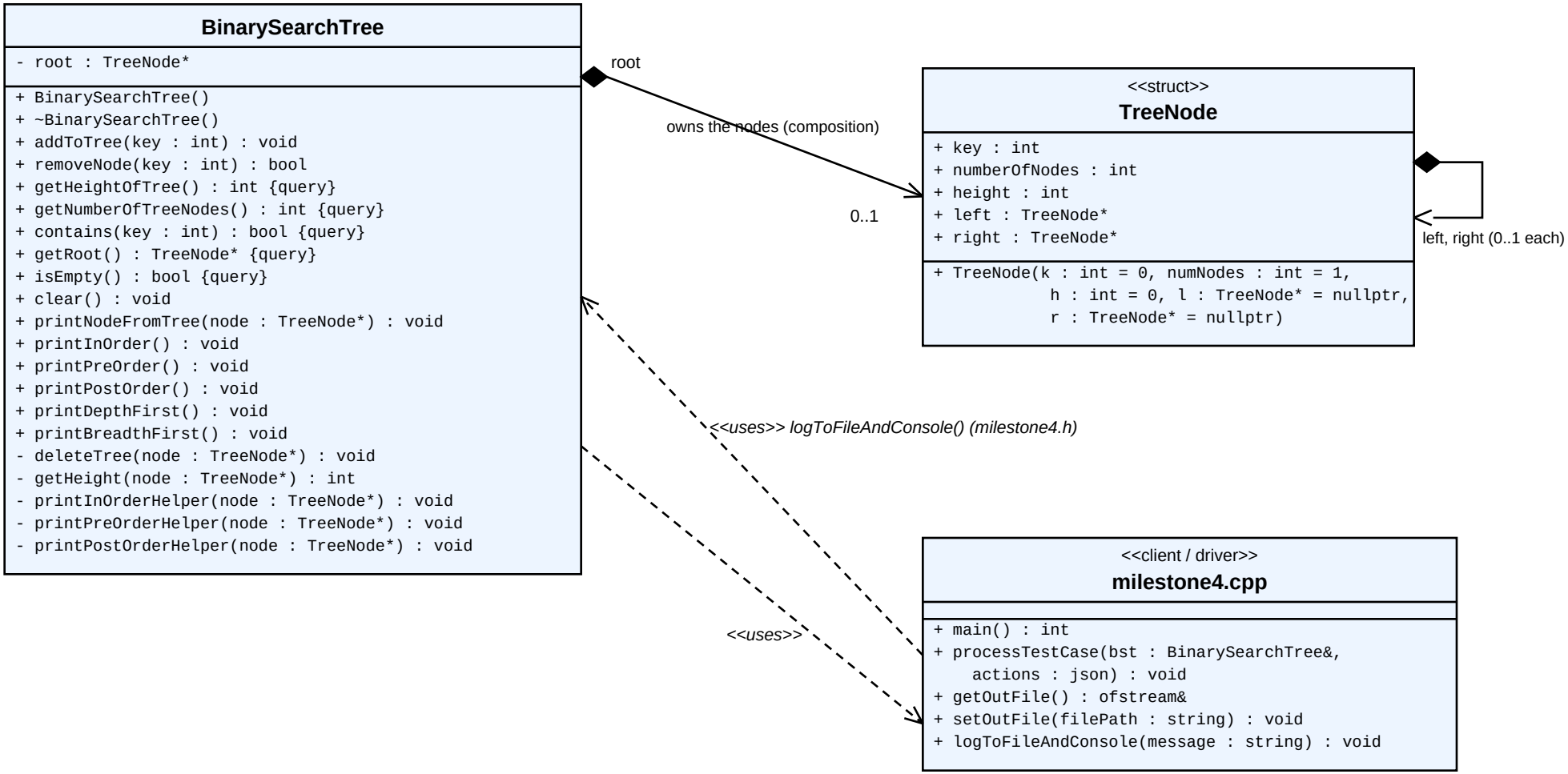

Notes:
- Height of an empty tree = 0, leaf = 1.
- Each node stores its own subtree height and node count.
- Duplicate keys are ignored.
- Traversal methods print through logToFileAndConsole().

UML Sequence Diagram (1 of 2)

addToTree(key) and removeNode(key) - called from processTestCase() in milestone4.cpp

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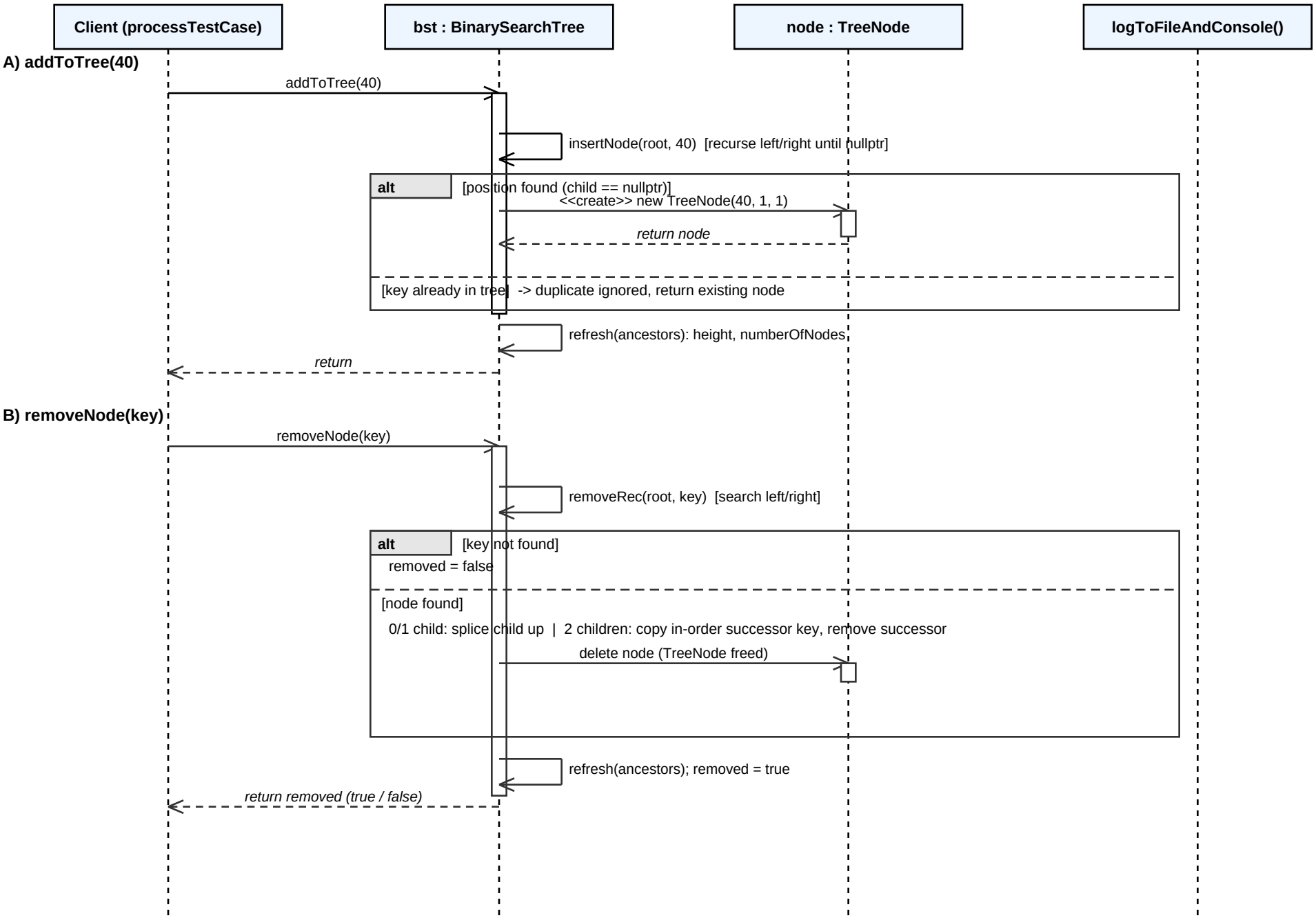

UML Sequence Diagram (2 of 2)

printBreadthFirst() and printInOrder() (pre/post-order are identical except for the order of the log call)

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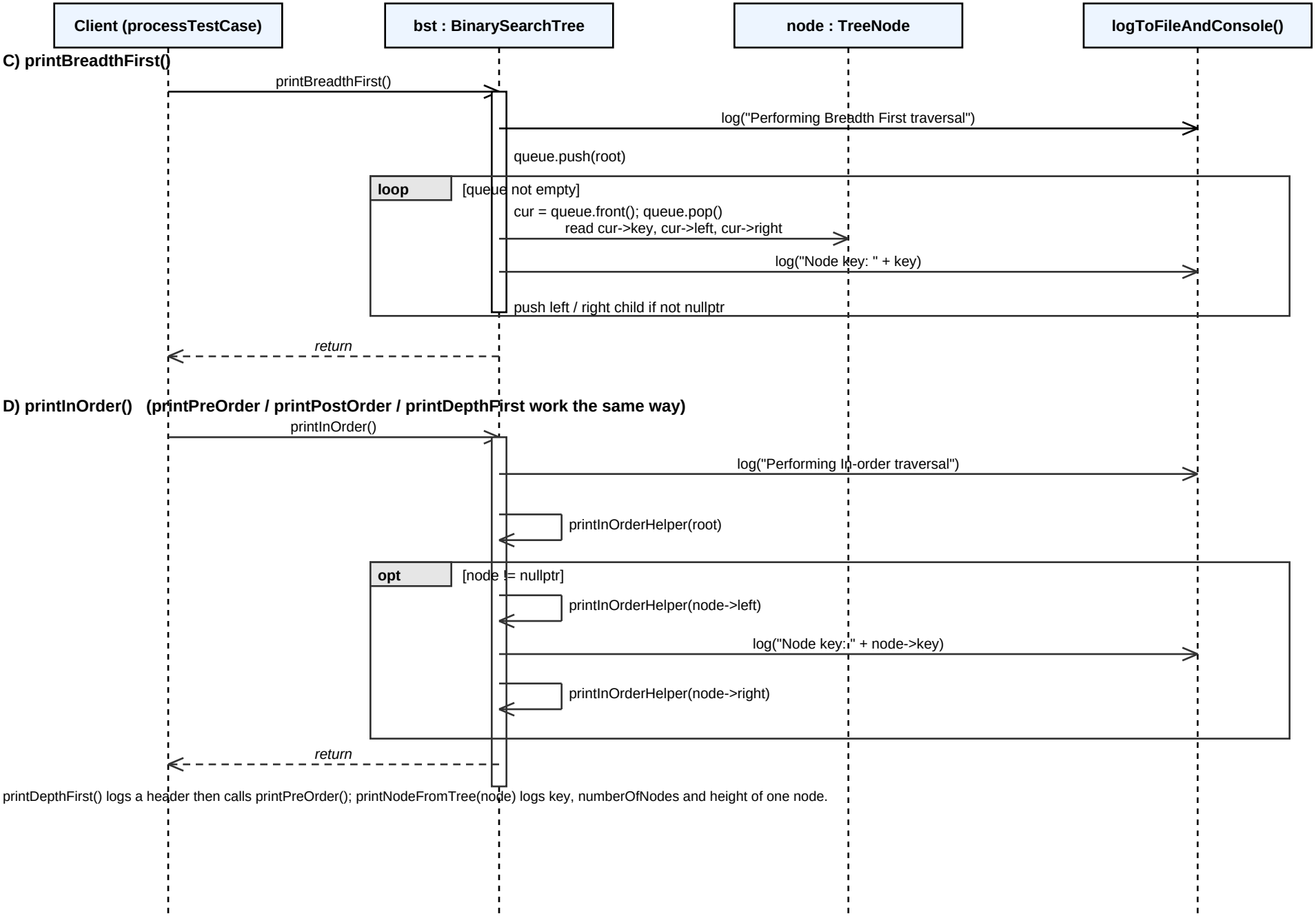

printDepthFirst() logs a header then calls printPreOrder(); printNodeFromTree(node) logs key, numberOfNodes and height of one node.